Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

Problem Set 5

1. Show how the function $f(w_1, w_2, w_3) = \sum m(0,2,3,4,5,7)$ can be implemented using a 3-to-8 binary decoder and an OR gate.

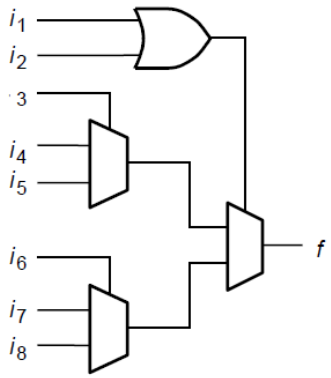
2. Consider the function $f = \overline{w}_1\overline{w}_3 + w_2\overline{w}_3 + \overline{w}_1 w_2$. Use the truth table to derive a circuit for f that uses a 2-to-1 multiplexer and any other necessary gates.

3. For the function $f(w_1, w_2, w_3) = \sum m(0,4,6,7)$ use Shannon's expansion to derive an implementation using a 2-to-1 Mux and any other necessary gates.

4. Consider the function $f = \overline{w}_2 + \overline{w}_1\overline{w}_3 + w_1 w_3$. Show how repeated application of Shannon's expansion can be used to derive the minterms of f.

5. Consider the function $f = \overline{w}_1\overline{w}_2 + \overline{w}_2\overline{w}_3 + w_1 w_2 w_3$. Give a circuit that implements f using the minimal number of two-inputs LUTs. Show the truth table implemented inside each LUT.

6. Actel Corporation manufactures an FPGA family called Act 1, which has the multiplexer-based logic block illustrated below. Show how the function $f = w_1\overline{w}_3 + \overline{w}_1 w_3 + w_2\overline{w}_3 + w_1\overline{w}_2$ can be realized using Act 1 logic blocks. Note that there are no NOT gates in the chip; hence complements of signals have to be generated using the multiplexers in the logic block.



7. Write a behavioural VHDL code that represents the function in Q1, using one selected signal assignment.

8. Using a conditional signal assignment, write a behavioural VHDL code for an 8-to-3 binary encoder.

9. Create a VHDL entity named *if2to4* that represents a 2-to-4 binary decoder using an if-then-else statement. Create a second entity named *h3to8* that represents the 3-to-8 binary decoder in Figure 6.17 of textbook, using two instances of the *if2to4* entity.

10. Design a shifter circuit, similar to the one in Figure 6.55 of textbook, which can shift a four-bit input vector, $W = w_3w_2w_1w_0$, one bit-position to the right when the control signal *Right* is equal to 1, and one bit-position to the left when the control signal *Left* is equal to 1. When *Right* = *Left* = 0, the output of the circuit should be the same as the input vector. Assume that the condition *Right* = *Left* = 1 will never occur.